

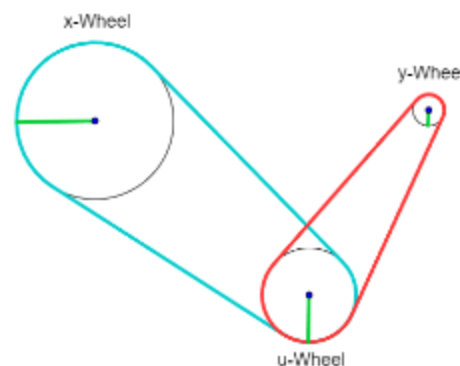
3.5 The Chain Rule

Motivation:

We wish to determine how to take the derivative of a composition of functions in multiple dimensions, i.e., what does chain rule look like for functions beyond $f(g(x))$?

Exploration:

Consider the mechanism to the right. Suppose that every time the x-wheel spins one time, the u-wheel spins 2 times. Furthermore, every time the u-wheel spins 1 time, the y-wheel spins 3 times. Question: If the x-wheel spins one time, how many times does the y-wheel spin? ¹



Recall: In calculus one you learned that if $y = f(x)$ and $x = g(t)$ where f and g are differentiable functions, then y is indirectly a differentiable function of t and $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$.

Note: We will see in this section that the chain rule generalizes nicely to functions of several variables.

Theorem (Case 1 of the Chain Rule):

Suppose that $z = f(x, y)$ is a differentiable function of x and y where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Proof: See pages 949 – 950 & Appendix D of the text.

Note: One may see the above equation written as: $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$.

¹ The corresponding applet can be found here:

https://webspace.ship.edu/msrenault/geogebra/calculus/derivative_intuitive_chain_rule.html

Example 1: Suppose that $z = \cos(x + 4y)$, $x = 5t^4$, and $y = \frac{1}{t}$. Assuming $t \neq 0$, find $\frac{dz}{dt}$.

Example 2: The pressure P (in kilopascals), volume V (in liters), and temperature T (in kelvins) of a mole of an ideal gas are related by $P = \frac{8.31T}{V}$. Find the rate at which the pressure is changing (rounded to the nearest thousandth) when the temperature is 300K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

Theorem (Case 2 of the Chain Rule):

Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(s, t)$ and $y = h(s, t)$ are differentiable functions of s and t . Then, $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$ and $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$.

Example 3: Suppose that $z = y(2^x)$ where $y = s^2 + st$ and $x = e^{st}$. Find the exact value of $\frac{\partial z}{\partial s}$ when $s = 2$ and $t = 0$.

Note: It is important to see that the symbol ∂z does not have a meaning in and of itself (unlike dz). Therefore, one should recognize that $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \neq \frac{\partial z}{\partial s} + \frac{\partial z}{\partial s}$ (unless it is the case that $\frac{\partial z}{\partial s} = 0$).

Theorem (General Chain Rule):

Suppose that u is a differentiable function of the n variables $x_1, x_2, \dots, x_n$ and each x_j is a differentiable function of the m variables $t_1, t_2, \dots, t_m$ and $\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$ for each $i = 1, 2, \dots, m$.

Proof: The proof the case 2 of the chain rule and the general chain rule are beyond the scope of the course.

Visualizing the General Chain Rule:

A great way to think through the general chain rule is with the help of a tree diagram. To make the idea of the tree diagram concrete, I will use a simple example.

Example 4: Suppose that $u = x^4y + y^2z^3$, $x = rse^t$, $y = rs^2e^{-t}$, and $z = r^2s \sin(t)$. Find the exact value of $\frac{\partial u}{\partial s}$ when $r = 2$, $s = 1$, and $t = 0$.